MOHAMMED ZAKWAN M

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TECHNICAL SKILLS

Core Programming: Python (Proficient), C, Java, SQL, JavaScript, HTML/CSS (Fundamentals)
Computer Vision & AI/ML: Object Detection (YOLO Architectures v1–v12, YOLO26), Object Tracking (ByteTrack), From-Scratch Model Implementation, Image Processing, Deep Learning Pipelines, Neural Networks
Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib, FastAPI
Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Antigravity
Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems

EDUCATION

Bachelor of Technology in Computer Engineering (Artificial Intelligence & Machine Learning) 2023 – 2027
Presidency University, Bangalore, Karnataka | CGPA: 7.24 / 10.0
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Storage Architecture

PROJECTS

Aerial & UAV Object Detection & Tracking System
Python, PyTorch, YOLO, OpenCV, TensorRT, ONNX | github.com/zakwanmunawar/AeroDrop-Tracker
- Trained a YOLO26n detection model (2.37M parameters) on a 12-class aerial dataset with 86,000+ annotations, reaching 22.1% mAP@50.
- Optimized the inference pipeline on an NVIDIA Tesla T4 GPU for sub-3ms latency, sustaining over 330 FPS.
- Built the preprocessing pipeline from scratch (frame resizing, bounding-box normalization, augmentation) and used OpenCV for live video inference with frame-to-frame tracking.

Agricultural Pest Surveillance System
Python, YOLO26n, ByteTrack, OpenCV, FastAPI, React, MQTT | github.com/zakwanmunawar/agrivision-pest-surveillance
- Fine-tuned YOLO26n on an insect dataset (80 epochs, 640 px input, AdamW, early stopping) for pest detection.
- Wrote an edge pipeline that runs inference on a live camera or video feed, assigns persistent IDs with ByteTrack, and publishes one MQTT event per tracked pest (type, confidence, timestamp, sensor ID).
- Built a FastAPI backend and React surveillance dashboard around the detection pipeline.

Construction Worker Safety & PPE Compliance Detector
Python, YOLO, OpenCV | github.com/zakwanmunawar/worker-safety-system
- Fine-tuned a 3M-parameter YOLO model to detect site safety violations, including falls and missing helmets or jackets, reaching 97.0% mAP@50 and 96.5% precision.
- Deployed the pipeline on a Tesla T4 GPU at sub-9ms latency, supporting real-time compliance alerts.
- Added color-coded bounding-box overlays in OpenCV to visually flag missing safety gear in the live feed.

Manhole Detection
Python, YOLOv8n, FastAPI, OpenCV | github.com/zakwanmunawar/Manhole-Detection
- Built a FastAPI service with an image-upload endpoint that runs pretrained YOLOv8n inference and returns the detection confidence with an annotated image.
- Geo-tags each report with latitude, longitude and foot-traffic density, and computes a repair-priority score weighting detection confidence and normalized foot traffic equally.

CERTIFICATIONS & ACHIEVEMENTS

YOLO Architecture Implementation & Research Series (v1–v12, YOLO26)
- Rebuilt YOLOv1 from scratch in PyTorch, then worked forward through later versions to understand what changed at each step.

Virtual Internship Certification in Python Programming – CodSoft 2025
- Completed a 4-week virtual internship, writing modular Python scripts and practicing core OOP and algorithmic concepts.

Data Structures & Algorithms in Python – NeetCode 75
- Worked through the NeetCode 75 set in Python, covering two pointers, sliding window, graphs, and dynamic programming.

Top 70 Selection (out of 500+ Teams) – University Innovation & Project Review (IPR)
- Built the hardware-software integration for an autonomous robotic car arm, one of 70 teams selected out of 500+.